

Rebecca Akeresola

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PROFESSIONAL PROFILE

- I am an emerging researcher with a PhD in Statistics, focusing on animal movement modelling for behavioural inference. My work lies at the interface of statistics and ecology, applying quantitative methods to analyse movement and environmental data in support of biodiversity conservation and evidence-based decision-making. My research interests include movement ecology, spatial statistics and the application of hidden Markov models to animal movement data.

RESEARCH EXPERIENCE

■ PhD Researcher — *School of Mathematics, University of Edinburgh (2021-2025)*

Supervisor: Prof. Víctor Elvira

Thesis: *Modelling animal movement for behavioural inference.*

- Developed and validated hidden Markov models for animal movement analysis
- Applied models to seabird visual tracking data to infer behaviour and inform conservation
- Investigated the impact of telemetry data resolution on behavioural inference
- Produced outputs through first author publication and 15+ presentation talks
- Collaborative engagement across:
 - University of Edinburgh (UoE)
 - Biomathematics and Statistics Scotland (BioSS)
 - Joint Nature Conservation Committee (JNCC)

■ MSc Researcher — *African Institute for Mathematical Sciences, Kigali, Rwanda (2019-2020)*

Supervisor: Dr. Ezra Gayawan

- Conducted Bayesian spatio-temporal modelling of greenhouse gas emissions across Africa
- Identified spatial and temporal trends relevant to policy and mitigation strategies

■ Undergraduate Researcher — *Dept. of Statistics, Federal University of Technology Akure (2017)*

Supervisor: Dr. Ezra Gayawan

- Conducted Bayesian spatial modelling of malnutrition among women in Nigeria
- Identified socio-economic determinants of regional health outcomes
- Produced outputs through first author publication

EDUCATION

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|-------------|---|
| 2020 – 2025 | ■ PhD (with integrated study), Statistics
University of Edinburgh, UK |
| 2019 – 2020 | ■ MSc, Mathematical Sciences (Distinction).
African Institute for Mathematical Sciences, Kigali, Rwanda |
| 2012 – 2017 | ■ B.Tech, Statistics (First Class Honours)
Federal University of Technology Akure, Nigeria |

PEER-REVIEWED PUBLICATIONS

- 1 **R. Akeresola**, A. Butler, E. Jones, R. King, V. Elvira, J. Black, and G. Robertson, "Validating hidden Markov models for seabird behavioural inference," *Ecology and evolution*, vol. 14, 2024, [Citations: 5, Full-text views: 2140].  DOI: 10.1002/ece3.11116.
- 2 **R. Akeresola** and E. Gayawan, "Analysis of the spatial patterns of malnutrition among women in Nigeria with a Bayesian Structured Additive Model," *Geojournal*, vol. 85, pp. 81–92, 2020, [Citations: 16, Full-text views: 371].  DOI: 10.1007/s10708-018-9958-0.

PRESENTATIONS

- Delivered 15+ research presentations across seminars, conferences, and academic meetings:
Invited talks: 2 | Conference presentations : 2 | Seminar presentations: 11

Selected Engagements

- Young Statisticians' Section of the Irish Statistical Association (Invited Talk, 2025)
- Joint Nature Conservation Committee (JNCC), Edinburgh (Invited Talk, 2023)
- National Centre for Statistical Ecology (NCSE) Meeting, Swansea (2023)
- Mathematical Biology Group Seminar, Edinburgh (2022, 2023, 2024)
- International Statistical Ecology Conference (ISEC), Swansea & South-Africa (2022, 2024)
- BioSS Annual General Meeting, Edinburgh (2022, 2023)
- MAC-MIGS CDT Symposium, Edinburgh (2021, 2022, 2023, 2024)

LEADERSHIP & ACADEMIC SERVICE

- **BioSS Representative — SEFARI Postgraduate Research Showcase, Edinburgh (2023)**
 - Selected to represent Biomathematics and Statistics Scotland (BioSS) at The Scottish Environment Food and Agriculture Research Institute (SEFARI) Showcase
 - Presented research to interdisciplinary audience
- **Session Chair — Centre for Statistics Annual Conference 2022, University of Edinburgh**
 - Chaired part of an academic session ensuring structured delivery and time management
 - Moderated discussion and facilitated Q&A

TEACHING & MENTORSHIP

- **Tutor — School of Mathematics, University of Edinburgh (2021–2024)**
 - Delivered tutorials across 6 semesters to undergraduate and MSc students (10-20 per class)
 - Provided student support and academic feedback for multiple courses
- **MSc Project Co-Supervisor — School of Mathematics, University of Edinburgh (Jul. – Aug. 2022)**
 - Co-supervised projects in Statistics with Data Science course
 - Guided research design, analysis, and interpretation

Professional Development

- **Bayesian Filtering Summer School**, Edinburgh, UK (2024)
- **Continuous-Time Modelling Workshop**, Edinburgh, UK (2023)
- **Animal Movement Modelling Online Summer School** (2023)
- **British Ecological Society Movement Ecology Conference** (Attendance), St. Andrews, UK (2023)

AWARDS, FUNDING & TRAINING

- **Funded Summer School on Animal Movement Modelling (2023)**
 - Funded by Graduate School for Veterinary Medicine and Animal Sciences, Swedish University of Agricultural Sciences
 - Selected participant for advanced training in animal movement modelling
- **EPSRC-funded PhD Studentship (2020-2024)**
 - Highly competitive funding awarded through MAC-MIGS Centre for Doctoral Training
- **African Institute for Mathematical Sciences (AIMS) Masters Scholarship (2019-2020)**
 - Highly competitive fully funded scholarship awarded by AIMS-Rwanda
 - Top candidates from across Africa chosen with < 6% selection rate.

PROFESSIONAL MEMBERSHIP

Member ■ Royal Statistical Society | Membership No: 211473

RESEARCH & TECHNICAL SKILLS

Quantitative	■ Statistical modelling, Movement modelling, Spatio-temporal modelling
Data analysis	■ Large scale data handling, high-resolution movement data analysis, reproducible workflows (RMarkdown)
Programming	■ R (advanced), Python (intermediate), $\text{\LaTeX}$ (scientific writing and publishing)

REFEREES

Prof Ruth King

Director of Bayes Centre
Thomas Bayes' Chair of Statistics
School of Mathematics, University of Edinburgh,
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Dr Adam Butler

Stakeholder Management and Impact Lead for
Offshore Renewables
Biomathematics and Statistics Scotland (BioSS),
Edinburgh
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Prof Víctor Elvira

Personal Chair of Statistics and Data Science
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